

Heredity

Accumulated variations, Mendel's experiments with pea plants, rules for inheritance of traits, and how sex is determined in human beings.

CBSE 2026-27 assessment note: Heredity is listed for formative/internal assessment and is not included in the year-end board examination.

Introduction

Children look like their parents but not exactly. Some features are similar. Some are different. The transmission of characters from parents to offspring is called heredity. The differences among individuals are called variations.

In the last chapter we saw that DNA copying brings small variations. In this chapter we study how those variations are inherited and what rules control the inheritance of traits.

Variations and Their Importance

Variation means a difference in a trait among individuals of the same species. For example, height, eye colour, or shape of ear lobe can vary.

Variations arise mainly due to two reasons. First, errors during DNA copying. Second, sexual reproduction where genes from two parents combine.

The effect of a variation depends on the environment: it may be advantageous, neutral or disadvantageous. Variations that improve survival or reproduction in a particular environment can become more common over many generations. Variation is the raw material for evolution.

Key terms

Heredity: transmission of characters from parents to offspring.

Variation: differences in characters among individuals of the same species.

Gene: unit of inheritance. A section of DNA that controls one trait.

Trait: a feature of an organism, such as plant height or eye colour. Controlled by genes.

Mendel and His Pea Experiments

Gregor Johann Mendel is called the Father of Genetics. He worked on garden pea plants (*Pisum sativum*) in the 19th century. He chose pea plants because they had many clear contrasting traits, a short life cycle, and could be self or cross-pollinated easily.

Mendel studied seven pairs of contrasting traits in pea. Two well known pairs are:

- Tall plant versus dwarf plant
- Round seeds versus wrinkled seeds
- Purple flowers versus white flowers
- Green seeds versus yellow seeds

Mendel crossed plants with contrasting traits and observed the results in the next generations. He kept careful counts. From these counts he gave the laws of inheritance.

Monohybrid Cross and Dominance

A monohybrid cross studies one trait at a time. Example: height of pea plant.

Experiment. Mendel crossed a pure tall plant (TT) with a pure dwarf plant (tt). The first generation (F1) had all tall plants. No dwarf plants appeared.

Then he self-pollinated the F1 tall plants. In the second generation (F2), he got both tall and dwarf plants in the ratio 3 tall : 1 dwarf. Out of 4 plants, 3 were tall and 1 was dwarf.

Explanation

Each trait is controlled by two factors (now called genes or alleles). One comes from the father and one from the mother. In Tt, T is dominant. It masks the effect of t. So Tt looks tall. Only tt looks dwarf. T is the dominant allele. t is the recessive allele.

Genotype and phenotype. Genotype is the gene combination, for example TT, Tt, or tt. Phenotype is the visible trait, for example tall or dwarf. In F₂, phenotype ratio = 3:1, genotype ratio = 1 TT : 2 Tt : 1 tt.

Law of dominance: in a heterozygous condition (Tt), the dominant trait is expressed and the recessive trait remains hidden.

Law of segregation: the two alleles for a trait separate during gamete formation. Each gamete gets only one allele. They come together again at fertilisation.

Dihybrid Cross and Independent Inheritance

A dihybrid cross studies two traits together. Mendel crossed pea plants differing in two traits: seed shape and seed colour.

Experiment. He crossed a plant with round and yellow seeds (RRYY) with a plant with wrinkled and green seeds (rryy). Round is dominant over wrinkled. Yellow is dominant over green.

F₁ generation: all plants had round and yellow seeds (RrYy).

He self-pollinated the F₁ plants. F₂ showed four types: 9 round yellow, 3 round green, 3 wrinkled yellow, 1 wrinkled green. Ratio = 9:3:3:1.

This result shows that the two traits are inherited independently. The inheritance of seed shape does not affect the inheritance of seed colour.

Law of independent assortment

Factors for different traits separate and assort independently during gamete formation. So a new combination of traits can appear in the offspring.

This explains the new combinations seen in Mendel's experiment. Independent assortment applies to the gene pairs used in such a cross; genes that are linked on the same chromosome may not assort independently.

How Traits are Inherited

Traits are controlled by genes present on chromosomes. Each cell has two copies of each chromosome, one from the father and one from the mother. So there are two copies of each gene.

Gametes (sperm and egg) are formed by meiosis. They have only one set of chromosomes. So each gamete carries only one allele for each trait. When sperm and egg fuse, the offspring gets two alleles again, one from each parent.

Some traits are dominant and some are recessive. The expressed trait depends on which alleles are present.

Acquired traits are not inherited. If a person cuts his hair or builds strong muscles by exercise, these changes are not passed to children. Only changes in the DNA of reproductive cells can be inherited. Changes in non-reproductive body cells do not go to next generation.

Sex Determination in Human Beings

Sex determination is the process that decides whether a child will be a boy or a girl.

Chromosomes in humans. Each human cell has 46 chromosomes. That is 23 pairs. Out of these, 22 pairs are autosomes which control body traits. The last pair is the sex chromosomes.

- In a female, the sex chromosomes are XX. So every egg carries one X chromosome.
- In a male, the sex chromosomes are XY. So a sperm can carry either an X or a Y chromosome.

How it works. When a sperm carrying X fuses with an egg (X), the child is XX, a girl. When a sperm carrying Y fuses with an egg (X), the child is XY, a boy. There is a 50 percent chance of each.

This means the X- or Y-bearing sperm determines the chromosomal sex of the child. The egg always contributes an X chromosome. Neither parent controls which sperm fertilises the egg, so blaming either parent is scientifically wrong.

Sex determination in other organisms. In many animals, sex is determined by similar XY or ZW systems. In some reptiles, environmental factors like temperature can influence sex determination. But in human beings, it is purely genetic.

Summary

- Heredity is transmission of traits from parents to offspring. Variations are differences among individuals.
- Variations arise due to errors in DNA copying and due to sexual reproduction. Useful variations accumulate over generations.
- Mendel studied seven traits in pea. He is the Father of Genetics.
- In a monohybrid cross, F₂ phenotype ratio is 3:1. The trait seen in F₁ is dominant. The hidden trait is recessive.
- Law of dominance and law of segregation explain monohybrid inheritance.
- In a dihybrid cross, F₂ ratio is 9:3:3:1. This proves law of independent assortment.
- Traits are controlled by genes on chromosomes. Gametes carry one allele. Zygote gets two.
- Acquired traits are not inherited. Only germ cell DNA changes are inherited.
- In humans, females are XX and males are XY. Sex of the child is decided by whether the sperm carries X or Y. Chance of boy or girl is 50 percent.

Practice Questions

1. Why do offspring show variations even though they inherit traits from their parents?

2. A pure tall plant (TT) is crossed with a pure dwarf plant (tt). What will be the phenotype and genotype of F₁ and F₂? Give ratios.

3. Explain the dihybrid cross with the example of round yellow and wrinkled green seeds. What does the 9:3:3:1 ratio prove?

4. Why are acquired traits not inherited? Give an example.

5. Explain chromosomal sex determination in human beings. Which chromosome can differ among sperm cells?

6. Why was the recessive trait not seen in the F1 generation but seen again in the F2 generation of Mendel's monohybrid cross?